

Clean & Transform dataset into Market basket analysis format

Cleaning:

First we have to check and remove negative quantity, cancelled Billno and drop missing Itemname

```
[29]: # For negative quantity:
df1 = df1[df1['Quantity'] > 0]

# For cancelled BillNo:
df1 = df1[~df1['BillNo'].str.startswith('C', na = False)]          ### ~ is used as a Not Operator

# For missing Itemname:
df1 = df1.dropna(subset = ['Itemname'])
```

Top 500 products:

```
[30]: top_products = df1.groupby('Itemname')['Quantity'].sum().sort_values(ascending = False).head(500).index
```

```
[31]: df1 = df1[df1['Itemname'].isin(top_products)]
```

```
[32]: df1.info()
```

```

<class 'pandas.core.frame.DataFrame'>
Index: 234798 entries, 0 to 519333
Data columns (total 9 columns):
#   Column          Non-Null Count  Dtype
---  -
0   BillNo          234798 non-null object
1   Itemname        234798 non-null object
2   Quantity        234798 non-null float64
3   CustomerID      234798 non-null object
4   Country         234798 non-null object
5   Price           234798 non-null float64
6   Present_Date    234798 non-null datetime64[ns]
7   Revenue         234798 non-null float64
8   Month           234798 non-null period[M]
dtypes: datetime64[ns](1), float64(3), object(4), period[M](1)
memory usage: 17.9+ MB

```

Reducing Transaction Noise.

Single item basket are useless for market basket analysis

```
[33]: basket_size = df1.groupby('BillNo')['Itemname'].nunique()

valid_invoices = basket_size[basket_size > 1].index

df1 = df1[df1['BillNo'].isin(valid_invoices)]
```

```
[34]: df1.info()

<class 'pandas.core.frame.DataFrame'>
Index: 233073 entries, 0 to 519333
Data columns (total 9 columns):
#   Column      Non-Null Count  Dtype
---  -
0   BillNo      233073 non-null object
1   Itemname    233073 non-null object
2   Quantity    233073 non-null float64
3   CustomerID  233073 non-null object
4   Country     233073 non-null object
5   Price       233073 non-null float64
6   Present_Date 233073 non-null datetime64[ns]
7   Revenue     233073 non-null float64
8   Month       233073 non-null period[M]
```

memory usage: 17.8+ MB

Transaction–Product Binary Matrix Creation

using Groupby and unstack, unstack converts one level of row into columns and fill value = 0 means that the empty cell can be replaced with 0

```
[35]: basket = df1.groupby(['BillNo', 'Itemname'])['Quantity'].sum().unstack(fill_value = 0)
```

```
## Binary conversion --> Because market basket has nothing to do with quantity, it only checks that the product is purchased or not
```

```
basket = basket.applymap(lambda x : 1 if x > 0 else 0)
```

```
C:\Users\Hi\AppData\Local\Temp\ipykernel_11376\1618984716.py:4: FutureWarning: DataFrame.applymap has been deprecated. Use DataFrame.map instead.
```

```
    basket = basket.applymap(lambda x : 1 if x > 0 else 0)
```

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